WELCOME TO

CSE230: DISCRETE MATHEMATICS

COURSE INTRODUCTION

Discrete Mathematics is a foundational course aimed at building strong logical reasoning, mathematical modeling, and problem-solving skills—abilities that are essential for success in computer science and engineering. This course follows the first half of Kenneth H. Rosen's widely used textbook of the same name. Weekly reading of the assigned textbook sections is required, and students are encouraged to actively engage with the material. Alongside this, students should also utilize **practice problem-sets** to prepare for exams. As the name suggests, "practice problems" only become truly helpful through actual practice.

From the second week onward, ***Student Tutors*** (STs) will be available both online and on campus to support students throughout the course. Whether to check a solution, ask for hints, or discuss a tricky concept, the consultation hours with STs and course instructors are there to help—and students are strongly encouraged to make full use of the consultation hours of both the course instructors and the STs.

Over the semester, there will be ***at least three assignments***, each with a one-week deadline. The **average of all assignment scores will be included** in the final grade. There will also be ***a minimum of five quizzes***, conducted during class/lecture sessions. Only the **best N–2 quiz scores will count** toward the final grade.

Some selected topics (detailed at the end of this document) are marked as **"*self-study*"**. To encourage curiosity and independent learning, both the midterm and final exams will include **optional bonus questions** (worth up to 5 extra marks) based on these topics. These bonus marks can be added to your exam scores, **without exceeding the total**. Self study topics are to be explored by the students themselves and are not part of the lecture plan.

MARKS DISTRIBUTION

Attendance	90%+ attendance = 5 marks	**5**
Assignments	Average of 3 (Minimum)	**10**
Quizzes	Best N–2, out of 5 (Minimum)	**20**
Midterm Exam	Converted from 50	**30**
Final Exam	Converted from 50	**35**
Bonus	3+3.5 bonus from midterm+final (capped)	**6.5**
TOTAL		**100+6.5**

IMPORTANT LINKS

• Course Content Folder:	https://tiny.cc/cse230spring26
• Official Discord Server:	https://discord.gg/QJHHBRRUFX
• Book Link:	https://drive.google.com/file/d/1apBrWJrTiOLrPpY-z1PxW8QQ-i5Pp6wI/view?usp=drive_link

COURSE PLAN

Lecture Details:

LECTURE #	BOOK CHAPTER	TOPICS	LINKS (CLICK TO OPEN)	
			SLIDE	PRACTICE SHEET
0	1.1	• Introduction and Motivation for DM • The notion of logic and proof • Meaning of a proposition • Logical Connectives	Lecture 1	-
1	1.1, 1.3	• Conditional Statements • Converse, Inverse, and Contrapositives • Biconditional Statements • Truth Table of Compound Proposition	Lecture 2	-
2	1.3, 1.4	• Truth Table of Compound Proposition a. Tautology, Contradiction and Contingency b. Logical Equivalences • Universal and Existential Quantification	Lecture 3	Practice Sheet 1
3	1.7	• Proof Techniques a. Direct Proof (Construction) b. Indirect Proof (Contraposition) c. Proof by Contradiction	Lecture 4	-
4	1.7	• Proof Techniques [Cont.] a. Direct Proof (Construction) b. Indirect Proof (Contraposition) c. Proof by Contradiction	Lecture 4	-
5	5.1	• Mathematical Induction: a. For summation b. For divisibility	Lecture 5	Practice Sheet 2
6	2.1	• Set Terminologies a. Set and Set Notations b. Empty Set, Subsets c. Venn Diagrams d. Power set, Cartesian Product	Lecture 6	-
7	2.1	• Set Terminologies [Cont.] a. Set and Set Notations b. Empty Set, Subsets c. Venn Diagrams d. Power set, Cartesian Product	Lecture 6	-

8	2.2	• Set Operations: Union, Intersection, Difference • Set Identities • Inclusion-Exclusion Principle	Lecture 7	Practice Sheet 3
9	X	Review & Problem Solving		-
	- - -	**MIDTERMS**		
10	2.4	• Sequences and their Summations [AP & GP] • Recurrence Relations	Lecture 8	-
11	8.2	• Linear Homogeneous Recurrence Relation	Lecture 9	-
12	8.2	• Linear Homogeneous Recurrence Relation [Cont.]	Lecture 9	Practice Sheet 4
13	6.1	• Rules of counting a. Sum Rule b. Product Rule c. Subtraction Rule	Lecture 10	-
14	6.2	• The Pigeonhole Principle	Lecture 11	Practice Sheet 5
15	6.3, 6.5	• Permutation & Combination	Lecture 12	-
16	6.3, 6.5	• Permutation & Combination [Cont.]	Lecture 12	Practice Sheet 6
17	4.1	• Divisibility • Modular Arithmetic • Congruence Relation	Lecture 13	-
18	4.3	• Prime Numbers • Prime Factorization • Relative Prime Number • GCD LCM	Lecture 14	-
19	4.2	• Logic of Integer Representation • Base Conversion • Modular Exponentiation	Lecture 15	Practice Sheet 7
20	X	Review & Problem Solving	Combinatorics Review	-
	- - -	**FINALS**		

Self Study Topics:

Midterm:

1. **LOGIC PUZZLES** (CHAPTER : 1.2.5, Pages 20 to 22)
2. **SATISFIABILITY** (CHAPTER : 1.3.5, Pages 33 to 37)
3. **NESTED QUANTIFICATION** (CHAPTER : 1.5, Pages 60 to 72)
4. **PROVES BY COUNTER-EXAMPLES** (CHAPTER: 1.7.7, Page 93)

Final:

1. **LINEAR NON HOMOGENEOUS RECURRENCE** (CHAPTER : 8.2.3, Pages 547 to 550)
2. **CLASSICAL CRYPTOGRAPHY** (CHAPTER : 4.6.2, Pages 311 to 314)
3. **DIVISION RULE** (CHAPTER : 6.1.5, Pages 414 to 415)